# 🏭 Industrial Manufacturing & Production Control System (IMPCS)

Language: ANSI C
Architecture: Structured Enterprise Console System
Storage: Binary File-Based Persistence
Version: 1.0

## 📌 Executive Summary

The Industrial Manufacturing & Production Control System (IMPCS) is a structured industrial enterprise simulation platform developed entirely in ANSI C.

The system integrates core manufacturing operations including:

• Secure administrator authentication
• Raw material inventory management
• Finished product tracking
• Controlled production processing
• Production cost computation
• Statistical production analytics
• Reorder monitoring
• Timestamp-based audit logging

IMPCS models centralized industrial workflow using deterministic structured programming and binary file-based persistence without reliance on external database systems.

## 🏗️ System Architecture

IMPCS follows a layered logical architecture:

• Authentication Layer
• Raw Material Management Module
• Finished Product Management Module
• Production Processing Engine
• Production Cost Analytics Engine
• Audit Logging Layer
• Binary Persistence Layer

Although implemented within a single source file, functional responsibilities remain logically segmented and modular.

# 🔐 Authentication Module

The system ensures secure administrative access.

Features:

• Automatic admin initialization on first execution
• Password hashing using djb2 algorithm
• No plaintext credential storage
• Binary verification against `manufacturing_users.dat`

Default Credentials:

**Username: admin**
**Password: admin123**

Hashing Logic:

```
hash = ((hash << 5) + hash) + character
```

# 🏗️ Raw Material Management Module

Each raw material record contains:

• Raw Material ID
• Material Name
• Available Quantity
• Cost Per Unit
• Reorder Level

Functional Capabilities:

• Add raw material
• View raw material inventory
• Detect low stock condition

When quantity reaches or falls below reorder level, the system displays:

**REORDER REQUIRED**

Storage File: `raw_materials.dat`

This module ensures controlled consumption of production resources.

# 📦 Finished Product Management Module

Each finished product record contains:

• Product ID
• Product Name
• Available Quantity
• Production Cost Per Unit

Functional Capabilities:

• Add finished product
• Automatic quantity update during production

Storage File: `finished_products.dat`

This module tracks inventory of manufactured goods ready for distribution.

# ⚙️ Production Processing Engine

The production engine manages conversion of raw materials into finished products.

Each production record contains:

• Production ID
• Raw Material ID
• Raw Quantity Used
• Finished Product ID
• Produced Quantity
• Total Production Cost
• Timestamp

Operational Flow:

• Validate raw material availability
• Deduct raw material quantity
• Increase finished product quantity
• Compute production cost
• Record production transaction
• Persist production data

Cost Formula:

```
Total Cost = Raw Quantity Used × Cost Per Unit
```

The system ensures raw material sufficiency before executing production.

Storage File: `production_records.dat`

# 📊 Production Cost Analytics Engine

The analytics module processes production records and computes:

• Mean (Average Production Cost)
• Variance
• Standard Deviation

Formulas Used:

```
Mean = Σ Production Costs / N
Variance = Σ (Cost − Mean)² / N
Standard Deviation = √Variance
```

Purpose:

• Evaluate cost consistency
• Measure production variability
• Analyze manufacturing financial stability
• Support operational efficiency assessment

# 📝 Audit Logging

All major system activities are recorded in:

```
manufacturing_logs.txt
```

Logged Events:

• Admin login
• Raw material addition
• Finished product addition
• Production execution
• Analytics execution
• System exit

Each log entry includes timestamp metadata for traceability and accountability.

# 💾 Data Storage Design

Binary files used:

| File | Purpose |
|---|---|
| manufacturing_users.dat | Authentication records |
| raw_materials.dat | Raw material inventory |
| finished_products.dat | Finished goods inventory |
| production_records.dat | Production transactions |
| manufacturing_logs.txt | Audit logs |

Binary storage ensures:

• Efficient record alignment
• Faster read/write operations
• Reduced storage overhead
• Persistent industrial data integrity

# ⚙️ Compilation & Execution

Compile:

```
gcc manufacturing_system.c -o manufacturing -lm
```

Run:

```
./manufacturing
```

# 🧠 Engineering Highlights

• ANSI C structured programming
• Industrial workflow abstraction
• File-based production database simulation
• Real-time raw material deduction
• Dynamic finished goods update
• Integrated cost computation
• Statistical production analytics
• Deterministic memory handling
• Timestamp-based operational traceability

# 🎯 Project Significance

IMPCS demonstrates the capability to design and implement centralized manufacturing workflow using low-level systems programming.

It reflects:

• Industrial process modeling
• Resource consumption tracking
• Structured production data persistence
• Cost analytics integration
• Secure authentication handling
• Enterprise-grade manufacturing logic

# 🏁 Conclusion

The Industrial Manufacturing & Production Control System (IMPCS) is a complete industrial enterprise simulation platform implemented in pure ANSI C.

It integrates:

Authentication
Raw Material Management
Finished Product Tracking
Production Processing
Cost Analytics
Audit Logging

All within a structured and deterministic runtime architecture.